Semantic-Guided Flow Matching for Fast and Accurate Remote Sensing Image Super-Resolution

Zhicheng Gong, Fangzhou Yi, Ling Guan, Chunzhu Dong, and Hui Zeng

***Abstract*—Current diffusion-based super-resolution methods for remote sensing images face two critical limitations: 1) excessive computational demands due to iterative sampling and 2) semantic inconsistency in reconstructed images caused by stochastic denoising. To address these challenges, we propose SfmSR, a semantic-guided flow matching model that achieves fast (one–five steps) and accurate reconstruction through a novel two-stage framework. Stage 1 performs large-scale pretraining on diverse remote sensing data to learn robust multiscale representations, while Stage 2 introduces semantic-guided fine-tuning, where extracted high-level features dynamically regulate the flow matching process. This dual-phase approach reduces sampling steps by two orders of magnitude while maintaining stability through deterministic ordinary differential equation (ODE)-based generation. Experimental results demonstrate that SfmSR outperforms existing methods on remote sensing image datasets such as Potsdam and Toronto. Moreover, SfmSR exhibits significant advantages in model complexity and inference efficiency, achieving fast inference with fewer parameters and lower memory usage, thus meeting real-time requirements in practical applications. Compared with state-of-the-art (SOTA) methods, SfmSR not only achieves superior reconstruction quality but also demonstrates clear advantages in inference speed and computational efficiency.**

***Index Terms*—Diffusion models, remote sensing, super-resolution, transformers.**

I. Introduction

HIGH-RESOLUTION remote sensing imagery (HRRSI) is a critical tool across numerous disciplines due to its ability to provide detailed ground object information and precise spatial resolution [1], [2], [3], [4], [5], [6], [7]. It offers robust data support for scientific research, resource management, and decision-making processes [3], [8], [9], [10]. For example, in land resource surveys, HRRSI enables accurate identification of land cover types and facilitates dynamic monitoring of land use changes [11]. In precision agriculture, it supports real-time crop growth monitoring [12], yield prediction [13], and agricultural planning [14]. Similarly, in ecological monitoring [15], HRRSI can extract critical information such as forest types, areas, and volumes, providing essential data for forest resource management and conservation efforts. However, acquiring high-resolution remote sensing images faces challenges, including high sensor costs, weather and revisit cycle limitations, and large data storage demands.

Over the past few years, deep learning has significantly improved image super-resolution reconstruction. Early models such as SRCNN [16] outperformed traditional methods, while subsequent architectures such as VDSR [17] and EDSR [18] enhanced reconstruction quality through deeper networks and residual learning. Recently, transformer-based models such as TransENet [19] improved feature representation by integrating multiscale and high-dimensional information. In addition, GAN-based models such as SRGAN [20], ESRGAN [21], and Real-ESRGAN [22] have achieved remarkable results in generating realistic textures and fine details through adversarial learning. Despite these advancements, challenges remain in improving reconstruction details and visual quality.

Recently, diffusion models, a novel class of deep generative models, have shown exceptional performance in tasks such as image generation [23], [24], [25], [26], [27], video synthesis [28], [29], [30], and natural language processing [31]. In image super-resolution [32], [33], [34], diffusion models generate high-resolution images through a gradual denoising process, significantly enhancing spatial details and object recognition capabilities. However, practical applications of diffusion models face several limitations [35]. First, their sampling process requires numerous iterative steps, resulting in high computational costs and slow generation speeds, which hinder real-time applications. Second, the stochastic nature of diffusion models can introduce instability, particularly in complex scenes where artifacts or detail distortions may occur. Finally, diffusion models may exhibit hallucination phenomena [35], generating structures or textures that do not align with real-world data, thereby reducing their reliability in practical scenarios. Addressing these challenges—such as improving sampling efficiency, ensuring generation stability, and mitigating hallucination effects—remains a critical focus for future research.

To address the limitations of the existing diffusion models—such as the need for thousands of sampling iterations, low accuracy, and hallucination phenomena caused by the stochastic nature of the diffusion process—we apply the

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theory of flow matching [36] to the task of single remote sensing image super-resolution reconstruction. We propose a novel semantic-guided flow matching model (SfmSR) that achieves accurate sampling results in just one–five steps, as illustrated in Fig. 1, significantly improving both the speed and precision of super-resolution reconstruction. Leveraging a large-scale remote sensing dataset, we pretrain the model without semantic conditions. This approach allows the model to learn reconstruction information across multiple scales and environments, enhancing its robustness and zero-shot capabilities. In the subsequent training phase, semantic information is introduced to address the potential loss of the initial input data distribution during sampling. By incorporating a deterministic diffusion process, only a first-order Euler sampler is required during inference, enabling high-resolution results to be obtained with very few sampling steps. The experimental results demonstrate that our proposed method outperforms current diffusion-based approaches, requiring fewer sampling steps and achieving faster inference speeds. The main contributions of this work are as follows.[1]

1) We propose a flow-matching-based algorithm for remote sensing super-resolution reconstruction, achieving superior image reconstruction quality with fewer sampling steps.
2) We introduce a novel 17M-parameter flow matching network that takes low-resolution images and their corresponding semantic-level representations as inputs. Compared with previous methods, this approach significantly reduces the number of parameters, improves inference speed, and enhances prediction accuracy.
3) The experimental results demonstrate that our proposed model outperforms current state-of-the-art (SOTA) diffusion-based super-resolution reconstruction models in terms of both pixel-level error and visual authenticity. In addition, we investigate various flow matching training strategies, offering insights and a foundation for future research in flow matching for remote sensing SR techniques.

II. Related Work

A. Nondiffusion Model-Based SR Methods

In remote sensing image super-resolution, traditional methods such as bilinear and bicubic interpolation and sparse representation struggled with high-frequency details. Deep learning shifted the paradigm, with SRCNN [16] pioneering CNN-based methods by learning low-to-high-resolution mappings, significantly improving reconstruction. VDSR [17] and EDSR [18] further enhanced accuracy through deeper networks and residual learning, while RDBPN [37] and WAB [38] improved feature representation. However, CNN-based methods often produce blurry outputs due to their average mapping approach.

GANs, such as SRGAN [20] and ESRGAN [21], introduced realistic textures through adversarial learning, with TE-SRGAN [20] addressing edge blur and SpaceGAN [39] combining RGB and hyperspectral data for large-factor reconstruction. Despite balancing pixel-level accuracy and visual quality, GANs face training instability.

Transformer-based methods such as SwinIR [40] and ESRT [41] excel in capturing long-range dependencies and global context, while recent work such as MambaHSISR [42] further advances hyperspectral image super-resolution by leveraging Mamba's linear-complexity modeling of spatial–spectral dependencies. SwinIR introduced hierarchical feature extraction, and ESRT improved efficiency through spatial and channel attention, yet these methods demand significant computational resources. In contrast, MambaHSISR addresses this by decomposing the problem into dedicated spatial and spectral subnetworks, achieving superior restoration quality while maintaining efficiency.

The existing methods often produce overly smooth outputs due to $L2/L1$ loss, misaligning with human perception. Our method overcomes this by learning a conditional vector field, achieving superior visual quality and detail preservation while maintaining computational efficiency, making it more practical for remote sensing applications.

B. Diffusion-Model-Based SR Methods

Diffusion models [27], an emerging class of generative models, have gained attention for their ability to generate high-quality, detail-rich images through gradual denoising. They excel in restoring complex textures and object details in remote sensing imagery. Score-based methods [43] use Langevin dynamics for image generation, while denoising diffusion probabilistic models (DDPMs) [44] introduced a Markov-chain-based framework. Denoising diffusion implicit models (DDIMs) [45] improved efficiency via non-Markovian sampling, and latent diffusion models (LDMs) [23] reduced complexity by operating in latent space. Song et al. [46] unified these methods into a stochastic differential equation (SDE) framework, and EDM [47] optimized sampling parameters by decoupling design choices.

In image super-resolution, diffusion models show promise, though their application in remote sensing remains exploratory. Studies like HSR-Diff leverage multispectral [48] data to enhance reconstruction accuracy, while ResDiff [49] combines CNNs with diffusion models for scalability. However, challenges persist when applying these methods to remote sensing images with rich spectral and shape features.

In single-image remote sensing super-resolution (SRSISR), Liu et al. [72] pioneered the integration of diffusion models. EHC-DMSR [50] used transformers and CNNs to extract conditional information and applied Fourier constraints for high-frequency detail. TESR [51] used a two-stage approach, and EDiffSR [34] introduced a conditional prior enhancement module (CPEM) and efficient activation network (EANet) to reduce computational costs. FastDiffSR [33] improved speed and accuracy through a novel sampling strategy.

Most existing methods rely on SDEs, which introduce uncertainty in the forward process, limiting sampling efficiency. Our method uses flow matching, categorized as an ordinary differential equation (ODE), to avoid these issues, enhancing training stability and inference efficiency.

[1]The code for this work is available at https://github.com/YuukiGong/SfmSR

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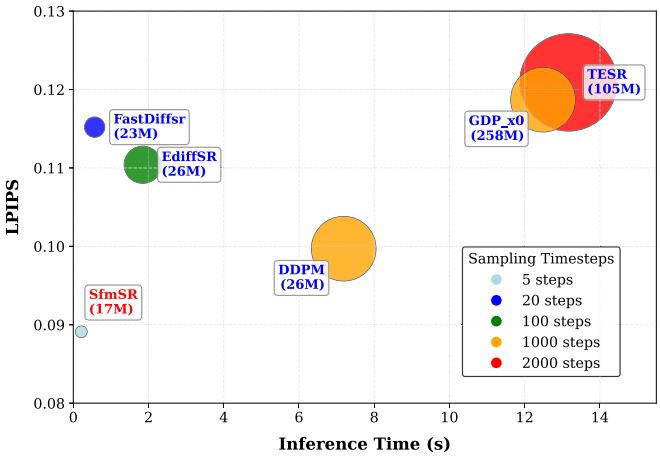

Fig. 1. Graph compares various super-resolution diffusion-based methods. Our proposed method (SfmSR) achieves low inference time, high perceptual quality (low LPIPS), and efficient parameter usage, outperforming other diffusion-based methods on the Potsdam dataset.

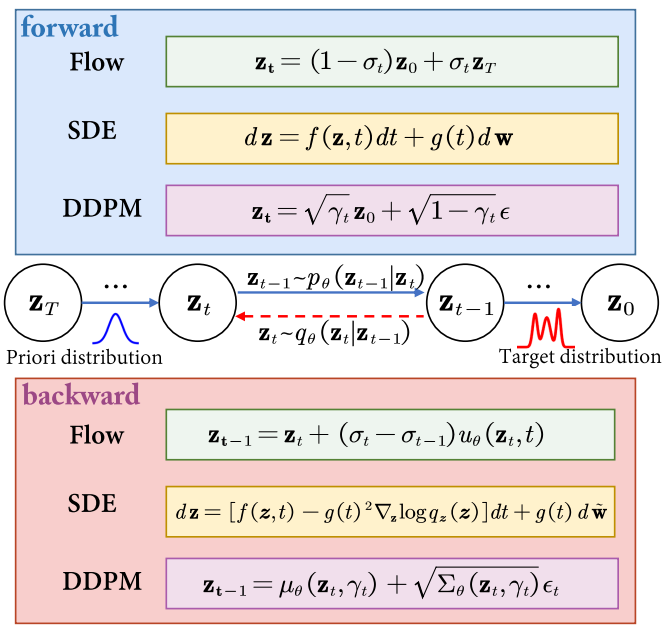

Fig. 2. Distribution-shifting processes in super-resolution with different generative approaches. Our proposed flow-based SfmSR method (highlighted in red) achieves superior performance with low inference time, high perceptual quality (low LPIPS), and efficient parameter usage on the Potsdam dataset.

III. Preliminaries

The probabilistic diffusion framework consists of two fundamental processes: a forward process that gradually adds noise to data, and a reverse process that learns to denoise and reconstruct the original data distribution. The differences between various diffusion models lie in their design of the forward and reverse processes, as illustrated in Fig. 2. In this article, we first discuss the differences between flow matching-based diffusion processes and SDE-based diffusion processes, followed by an exploration of our proposed semantic-guided flow matching reconstruction method.

A. Discrete-Time Formulation

The forward process is defined as a Markov chain that gradually transforms the data distribution $q(z_0)$ into a simple noise distribution through T steps, which is first proposed in [44]

$$q(z_t|z_{t-1}) = \mathcal{N}\left(z_t \mid \sqrt{1-\alpha_t}z_{t-1}, \alpha_t \mathbf{I}\right) \tag{1}$$

where
z_t noisy data at time step t
α_t controls the noise variance schedule;
$\mathbf{I}$ identity matrix.

According to [44], an important property of this process is that it permits direct sampling at any timestep t from the initial state z_0

$$z_t = \sqrt{\gamma_t}z_0 + \sqrt{1-\gamma_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}) \tag{2}$$

where $\gamma_t = \prod_{s=1}^{t}(1-\alpha_s)$ represents the cumulative product of noise variances.

As shown in [44], the reverse process learns to invert this noise addition

$$p_\theta(z_{t-1}|z_t) = \mathcal{N}(z_{t-1} \mid \mu_\theta(z_t, t), \Sigma_\theta(z_t, t)) \tag{3}$$

where μ_θ and Σ_θ are parameterized by a neural network trained to minimize the variational lower bound on the data likelihood.

B. Continuous-Time SDE Formulation

The discrete-time process can be extended to continuous time through the following derivation from [46], where the discrete steps are generalized to an SDE framework:

1) *Time Continuization:* We transition to $t \in [0, 1]$ by defining the below.
   a) $\alpha_t = \beta(t)\Delta t$ (continuous noise schedule).
   b) $\gamma_t = \gamma(t)$ (cumulative noise schedule).
2) *Discrete Approximation:* The forward process becomes

$$z_t \approx \left(1 - \frac{1}{2}\beta(t)\Delta t\right)z_{t-1} + \sqrt{\beta(t)\Delta t}\,\epsilon_t. \tag{4}$$

3) *SDE Formulation:* Taking the limit $\Delta t \to 0$ yields

$$dz_t = -\frac{1}{2}\beta(t)z_t dt + \sqrt{\beta(t)}d\mathbf{w}_t. \tag{5}$$

The resulting SDE consists of the below.
1) A deterministic drift term $(-(1/2)\beta(t)z_t dt)$.
2) A stochastic diffusion term $((\beta(t))^{1/2}d\mathbf{w}_t)$.

C. Challenges and Our Approach

While powerful, this framework presents several challenges.
1) The reverse SDE is computationally intensive to solve.
2) The entropy-increasing forward process is fundamentally irreversible.
3) Multiple diffusion paths create prediction uncertainty [36].

To address these limitations, we propose a flow matching approach that does the following.
1) Replaces the computationally intensive SDE with a flow-based ODE formulation, reducing sampling steps

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by 20× while maintaining reconstruction accuracy for remote sensing imagery.

2) Establishes a well-posed prior distribution and eliminates forward process uncertainty through exact vector field learning, particularly crucial for preserving geometric features in aerial/satellite images.
3) Improves training stability through exact likelihood computation while maintaining sample quality and reducing computational complexity.

D. Flow Matching Framework

Flow matching [36] is a generative modeling technique that matches probability flows to achieve sample generation. It defines a continuous-time evolution process $t \in [0, 1]$, transforming an initial distribution $p_0(z)$ into a target distribution. This process is governed by a time-dependent vector field $v_t(z)$, which determines the direction of sample evolution at time $t \in (0, 1)$. The evolution of a sample z_t is described by the ODE

$$\frac{dz_t}{dt} = v_t(z_t) \tag{6}$$

with the initial condition $z_0 \sim p_0(z)$. The goal is to define $v_t(z)$ such that the final distribution matches the target distribution. The continuity equation for the probability flow is

$$\frac{\partial p_t(z)}{\partial t} + \nabla \cdot (p_t(z) v_t(z)) = 0 \tag{7}$$

where $\nabla\cdot$ denotes the divergence operator. This equation ensures that the probability density evolves according to the vector field $v_t(z)$. The forward process, in its ODE form, is

$$dz_t = v_t(z_t)\, dt. \tag{8}$$

For a Gaussian prior, the evolution simplifies to

$$z_t = (1 - \sigma(t)) z_0 + \sigma(t) \epsilon \tag{9}$$

where ϵ remains constant for each t. This deterministic mapping, rather than a Gaussian distribution, enhances training efficiency and generation quality, making flow matching particularly suitable for tasks such as super-resolution.

IV. Methodology

The proposed SfmSR offers a significant advantage: it enables fast sampling, reducing the number of sampling steps from hundreds or even thousands to just one–five steps while maintaining superior sampling quality. The training of SfmSR is divided into two stages: 1) unconditional pretraining on a large-scale dataset and 2) semantic-guided fine-tuning. Each stage uses distinct training and sampling strategies.

A. Pretraining on Remote Sensing Images

Pretraining is a proven strategy in deep learning, enhancing performance in computer vision [52], NLP [53], and beyond [54]. For example, ImageNet [55] pretraining has boosted downstream vision tasks, while diffusion models trained on large datasets such as LAION-5B [56] show improved image quality and diversity. Inspired by these successes, we use large-scale remote sensing dataset pretraining for super-resolution tasks. Remote sensing images feature complex scenes and multiscale variations, making small datasets prone to overfitting. Pretraining on large datasets enables the model to learn rich spatial and contextual features, providing strong prior knowledge for super-resolution.

Unlike previous works that use U-Net as the backbone, SfmSR uses a transformer-based architecture [57], [58], as illustrated in Fig. 3. Pretraining optimizes the model parameters, enhances its ability to model long-range dependencies, and accelerates convergence. Therefore, drawing on successful practices from other domains, we use large-scale remote sensing dataset pretraining as the starting point of our training pipeline to improve the model's generalization and super-resolution performance. To enhance training speed and sampling effectiveness, we propose a novel sampling strategy for SfmSR. During training, we design three types of time sampling functions: linear, cosine, and exponential. The cosine-based sampling function is defined as follows:

$$P(N) = \frac{\cos\left(\frac{\pi}{2}\left(\frac{N}{M}\right)^{\alpha}\right)}{\sum_{i=0}^{M} \cos\left(\frac{\pi}{2}\left(\frac{i}{M}\right)^{\alpha}\right)} \tag{10}$$

where N is the time step, ranging from 0 to M, M is the maximum time step, and α is the decay rate control parameter. The exponential-based sampling function is defined as follows:

$$P(N) = \frac{\lambda^N}{\sum_{i=0}^{M} \lambda^i} \tag{11}$$

where λ is the decay rate. During pretraining, we use the exponential sampling function with $M = 1000$. If λ is too small, the probability of sampling earlier time steps increases; conversely, if λ is too large, the model tends to sample later time steps. The results of different sampling strategies are discussed in Section V-E.3.

B. Semantic Guidance Strategies

Flow matching recalculates the next inference result at each sampling step, gradually losing the initial input distribution over iterations. While this allows the model to experience varying clarity during pretraining, reintroducing the initial distribution during fine-tuning is essential for accuracy. The initial distribution acts as a conditional guide, ensuring stable inference and sampling. Initially, we used the original image's pixel information as a conditional guide. While this improved training convergence, the model failed to infer the correct vector field during inference. Let the downsampling operation be denoted as $D(\cdot)$ (e.g., bicubic interpolation), such that

$$y = x_0 = D(x_1) \tag{12}$$

where y is the condition (downsampled image), x_0 is the initial sample, and x_1 is the target sample. In our flow matching framework, the intermediate sample x_t is an interpolation

$$x_t = (1 - t) x_0 + t x_1. \tag{13}$$

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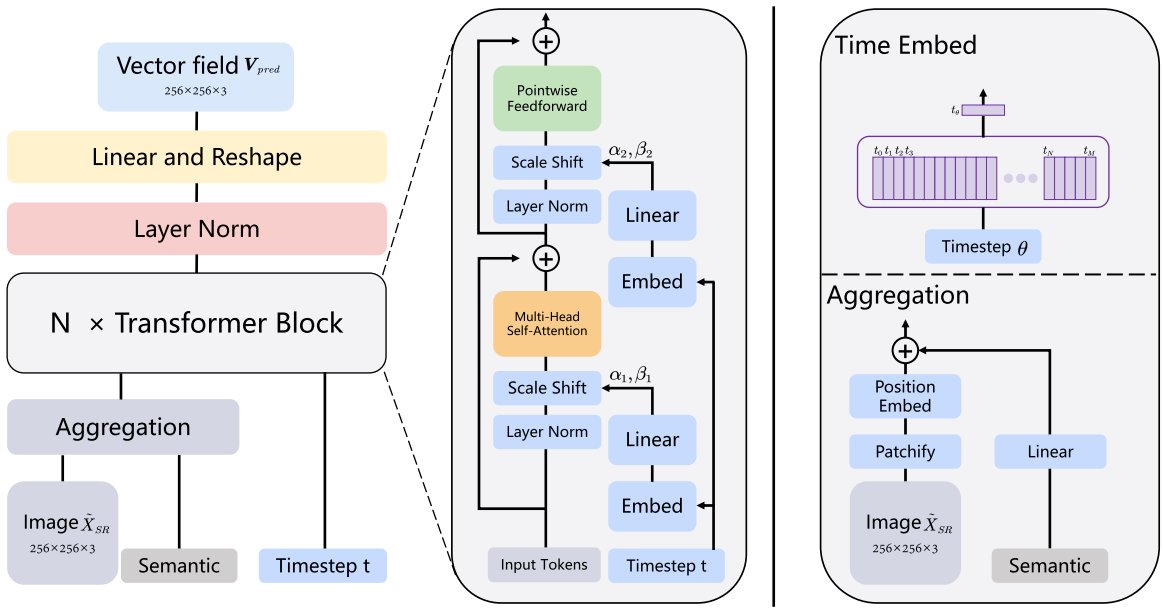

Fig. 3. Architecture of the network in our SfmSR.

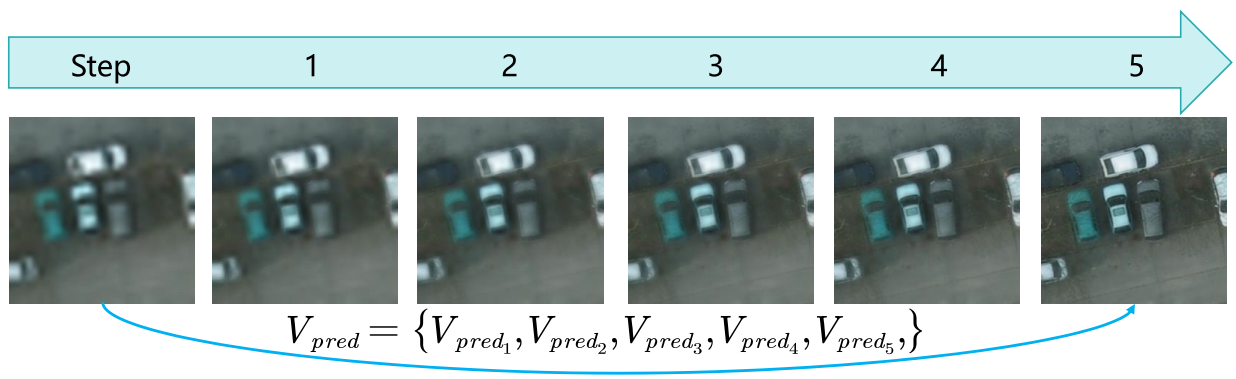

Fig. 4. Schematic of the flow matching process during the inference stage. Since the vector fields predicted at different noise levels are approximately the same, the final result can be obtained through one-step sampling.

Feeding y directly into the network, the predicted vector field $u_t(x_t|y)$ should satisfy

$$u_t(x_t|y) = \frac{dz_t}{dt} = x_1 - x_0. \tag{14}$$

Since $y = x_0$, we have

$$x_t - y = t(x_1 - x_0) = x_t - x_0. \tag{15}$$

Here, t, x_0, and x_t are known, so the training objective becomes $u_t(x_t|y) = (x_t - y/t)$. However, the network merely learns interpolation coefficients between the current and conditional samples, failing to estimate the correct conditional vector field from the output distribution.

To address this issue, we introduce semantic information from the original distribution to resolve the training failure caused by the mapping relationship between the conditional input and the intermediate sample in super-resolution tasks. Inspired by the findings of ScoreClip [59], [60], Clip -extracted semantic information aligns well with human perception. Therefore, we incorporate Clip semantic information into the network for conditional encoding. By extracting deep semantic features from the low-resolution image x_0 and aligning them with the features of the intermediate sample x_t, we enhance the network's ability to understand conditional information. This approach not only mitigates training difficulties caused by resolution mismatches but also better preserves the semantic consistency of the original image when generating high-resolution images. The sampling process is illustrated in Fig. 4. The experimental results show that introducing semantic information significantly improves the model's generation quality and training stability. To further improve the quality of generated images, we introduce Laplacian perceptual loss on top of the flow matching loss to capture high-frequency details in the image. This effectively enhances edge and texture details in the generated images. The final training objective combines the flow matching loss ($\mathcal{L}_{\text{FM}}$) and Laplacian perceptual loss ($\mathcal{L}_{\text{Laplacian}}$) with weight λ_l

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FM}} + \lambda_l \mathcal{L}_{\text{Laplacian}} \tag{16}$$

where $\mathcal{L}_{\text{FM}}$ is the flow matching loss

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{t,x_t}\left[\|u_t(x_t|y) - v_t(x_t)\|^2\right] \tag{17}$$

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE POTSDAM DATASET

Method	PSNR↑	FID↓	LPIPS↓	Params(M)
×2 Super-Resolution				
Bicubic	35.913	11.785	0.0922	–
SwinIR	36.112	10.892	0.0887	11.852
HSENet	36.208	10.785	0.0879	5.433
TransENet	36.305	10.632	0.0864	37.459
NDSRGAN	35.876	11.324	0.0901	17.510
HAT	36.287	10.702	0.0872	25.821
DDPM	36.142	10.825	0.0882	26.166
FastDiffsr	36.498	10.512	0.0837	23.229
Ours	**36.573**	**10.442**	**0.0815**	17.113
×4 Super-Resolution				
Bicubic	30.689	63.785	0.4432	–
SwinIR	32.919	44.590	0.2999	11.852
HSENet	32.932	44.192	0.3014	5.433
TransENet	32.768	41.310	0.2878	37.459
NDSRGAN	30.529	51.209	0.2137	17.510
HAT	32.987	44.932	0.2962	25.821
DDPM	31.989	24.307	0.0997	26.166
FastDiffsr	33.130	22.832	0.1152	23.229
Ours	**33.142**	**22.682**	**0.0891**	17.113

with $v_t(x_t)$ as the target vector field (14) and $u_t(x_t|y)$ as the network's conditional prediction, and $\mathcal{L}_{\text{Laplacian}}$ is the Laplacian perceptual loss

$$\mathcal{L}_{\text{Laplacian}} = \mathbb{E}_{x_1,\hat{x}_1}\left[\sum_{k=1}^{K}\|\mathcal{L}_k(x_1) - \mathcal{L}_k(\hat{x}_1)\|_1\right] \tag{18}$$

where $\mathcal{L}_k(\cdot)$ denotes the kth level feature of the Laplacian pyramid, x_1 is the target high-resolution image, $\hat{x}_1$ is the generated image, and K is the number of pyramid levels. By jointly optimizing $\mathcal{L}_{\text{FM}}$ and $\mathcal{L}_{\text{Laplacian}}$, our method achieves high-quality image generation while preserving semantic information and fine details. The experimental results show that $\mathcal{L}_{\text{Laplacian}}$ significantly improves generation quality.

V. EXPERIMENTS

A. Datasets

In our experiments, three datasets were used: the Functional Map of the World (fMoW) [61] dataset for pretraining, and the Potsdam and Toronto datasets [33] for fine-tuning and evaluation. The fMoW dataset provides high-resolution multispectral remote sensing images with diverse geographical scenes, enabling the model to learn rich spatial and multiscale features, enhancing its generalization and understanding of complex scenes. The Potsdam dataset, with 5-cm resolution, includes six annotated categories (e.g., buildings, vegetation), while the Toronto dataset, with 15-cm resolution, features urban scenes from North America (e.g., buildings, streets, cars). The training and testing splits were consistent with those used in previous methods.

B. Evaluation Metrics

To evaluate our method, we adopted three widely used metrics in super-resolution tasks: peak signal-to-noise ratio (PSNR) [62], Fréchet inception distance (FID) [63], and learned perceptual image patch similarity (LPIPS) [63]. PSNR measures pixel-level reconstruction accuracy, FID assesses visual authenticity by comparing feature distributions of generated and real images, and LPIPS quantifies perceptual similarity based on deep learning features, aligning with human visual perception. These metrics collectively evaluate numerical accuracy, visual quality, and perceptual similarity. For consistency with prior work, we followed the same experimental protocols and evaluation metrics as established in previous studies. In Table I, an upward arrow indicates that a larger value is preferable, while a downward indicates that a smaller value is better. The bold text denotes the optimal method for a given metric, and the underline signifies the second-best method.

C. Implementation Details

All THE methods were implemented in PyTorch, with experiments conducted on two NVIDIA RTX A6000 GPUs (48-GB memory). The proposed method was trained using the AdamW optimizer [64], with the number of sampling steps set to five. During pretraining, the learning rate was set to $2e^{-4}$ to ensure stable convergence and effective feature learning on the large-scale remote sensing dataset. For fine-tuning, the learning rate was set to $1e^{-4}$.

D. Experimental Results

In this work, we compare the proposed method with the baseline bicubic interpolation method and ten SOTA SR models in the remote sensing domain, including SwinIR [40], HSENet [65], TransENet [19], NDSRGAN [66], HAT [67], DDPM [44], TESR [51], GDP$_{x0}$ [68], EdiffSR [34], and FastDiffsr [33], in terms of quantitative metrics and inference speed.

1) Potsdam Datasets: The results are summarized in Table I, and visual quality comparisons are shown in Fig. 5. By comparing metrics such as PSNR, FID, LPIPS, and the number of parameters (Params), we comprehensively assess the performance of each model in remote sensing image super-resolution tasks.

In Table I, the proposed SfmSR method achieves optimal performance across both ×4 and ×2 super-resolution tasks. For ×4 SR, it attains the highest PSNR (33.142), lowest FID (22.682), and best LPIPS score (0.0891), showing 22% LPIPS improvement over the second-best method (FastDiffsr). Similarly for ×2 SR, SfmSR maintains superior performance with PSNR = 36.573, FID = 10.442, and LPIPS = 0.0815, outperforming FastDiffsr by 0.2 dB in PSNR and 1.3% in FID. Notably, SfmSR achieves these results with only 17.113M parameters—26% fewer than FastDiffsr (23.229M) and 93% fewer than GDP_x0 (258.844M), demonstrating consistent advantages in both reconstruction quality and computational efficiency across scaling factors.

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(a) (b) (c) (d) (e) (f)

Fig. 5. Visual comparisons of SR results with different SOTA methods on the Potsdam dataset. (a) Low resolution. (b) SwinIR. (c) HAT. (d) FastDiffSR. (e) SfmSR (ours). (f) Ground truth.

(a) (b) (c) (d) (e) (f)

Fig. 6. Visual comparisons of SR results with different SOTA methods on the Toronto dataset. (a) Low resolution. (b) SwinIR. (c) HAT. (d) FastDiffSR. (e) SfmSR (ours). (f) Ground truth.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE TORONTO DATASET

Method	PSNR↑	FID↓	LPIPS↓	Params(M)
×2 Super-Resolution				
Bicubic	31.021	24.819	0.0818	–
SwinIR	31.215	23.912	0.0802	11.852
HSENet	31.308	23.785	0.0794	5.433
TransENet	31.402	23.632	0.0779	37.459
NDSRGAN	31.123	24.324	0.0806	17.510
HAT	31.384	23.702	0.0787	25.821
FastDiffsr	31.398	22.512	0.0773	23.229
Ours	**31.413**	**21.733**	**0.0786**	17.113
×4 Super-Resolution				
Bicubic	24.722	107.589	0.4691	–
SwinIR	26.946	77.645	0.2285	11.852
HSENet	27.555	74.868	0.2319	5.433
TransENet	26.884	62.745	0.2260	37.459
NDSRGAN	25.653	44.778	0.1584	17.510
HAT	27.545	78.677	0.2214	25.821
DDPM	26.813	**38.377**	0.1022	26.166
TESR	26.983	41.259	0.1216	105.128
GDP_x0	26.958	51.146	0.1251	258.844
EdiffSR	26.170	54.313	0.0985	26.790
FastDiffsr	**27.901**	42.273	0.1011	23.229
Ours	27.621	39.712	**0.0972**	17.113

2) Toronto Datasets: We also compare the performance of SfmSR with other advanced models on the Toronto dataset. The results are summarized in Table II, and visual quality comparisons are shown in Fig. 6.

From Table II, the proposed SfmSR method demonstrates competitive results across both ×2 and ×4 super-resolution tasks. For ×2 SR, SfmSR achieves the best FID (21.733) and competitive PSNR (31.413), outperforming FastDiffsr by 3.5% in FID while maintaining similar PSNR. In ×4 SR, SfmSR achieves a PSNR of 27.621, slightly lower than FastDiffsr (27.901) but superior to other methods such as HSENet (27.555) and HAT (27.545), indicating strong reconstruction accuracy. On the FID metric, SfmSR achieves a score of 39.712, significantly lower than most methods (e.g., SwinIR at 77.645 and HSENet at 74.868) and close to DDPM's SOTA score of 38.377, highlighting its ability to generate visually authentic and diverse images. In addition, SfmSR attains the best LPIPS score of 0.0972, outperforming all other methods and further validating its superior perceptual quality.

3) Model Performance and Complexity Comparison: In this section, we compare the SfmSR network architecture with other advanced models in terms of model parameters (Params), memory usage (Memory), computational complexity (Gflops), and inference time (Inference Speed). The results are summarized in Table III. By evaluating these metrics,

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TABLE III
COMPARISON OF MODEL COMPLEXITY AND INFERENCE EFFICIENCY

TYPE	METHOD	MEMORY (MB)	FLOPS(G)	INFERENCE SPEED (S)	PARAMS(M)
NON-DIFFUSION-BASED	SWINIR	3298	808.74	0.2844	11.852
	HSENET	2630	279.90	0.2483	5.433
	TRANSENET	3980	342.98	0.1219	37.459
	NDSRGAN	2736	1230.50	0.1638	17.510
	HAT	4948	1739.25	0.3179	25.821
DIFFUSION-BASED	DDPM	10966	62.28	7.1895	26.166
	TESR	19866	179.33	13.1632	105.128
	GDP_X0	20808	843.05	12.4891	258.844
	EDIFFSR	2207	**46.25**	1.8462	26.790
	FASTDIFFSR	19186	175.17	0.5683	23.229
	OURS	**1231**	232.28	**0.2135**	**17.113**

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT TRAINING STRATEGIES ON THE POTSDAM DATASET

TRAINING STRATEGY	PSNR↑	FID↓	LPIPS↓
ZERO-SHOT	31.686	33.541	0.1542
WITHOUT PRE-TRAINING	33.108	22.734	0.0897
FINE-TUNING	**33.142**	**22.682**	**0.0891**

we comprehensively assess the computational efficiency and practical feasibility of each model.

From Table III, the proposed SfmSR demonstrates significant advantages in model complexity and inference efficiency. SfmSR requires only 1231 MB of memory, significantly lower than other diffusion-based models and even some nondiffusion models. By adopting a transformer-based architecture, SfmSR achieves faster sampling with fewer than five steps, trading higher Gflops for reduced computational overhead. With an inference speed of 0.21358 s, SfmSR outperforms all the diffusion-based models and remains competitive with nondiffusion models, making it suitable for real-time applications. Furthermore, SfmSR's parameter count (17.113M) is substantially lower than high-performance diffusion models and even some nondiffusion models, highlighting its efficient design.

E. Ablation Study

1) Detailed Results and Extended Discussions: In this section, we conduct a series of comparative experiments on the Potsdam dataset to investigate the impact of pretraining, different time sampling functions during training, the effect of semantic guidance, the influence of loss function weightings, and the results of varying sampling steps during inference.

2) Impact of Pretraining: Large-scale datasets have been proven to improve model performance in many domains, and we find that this also holds true for remote sensing image super-resolution tasks. To validate the effectiveness of pretraining, we compare the model's performance under three different training strategies on the Potsdam dataset: zero-shot transfer, training without pretraining, and fine-tuning after pretraining. The experimental results are shown in Table IV.

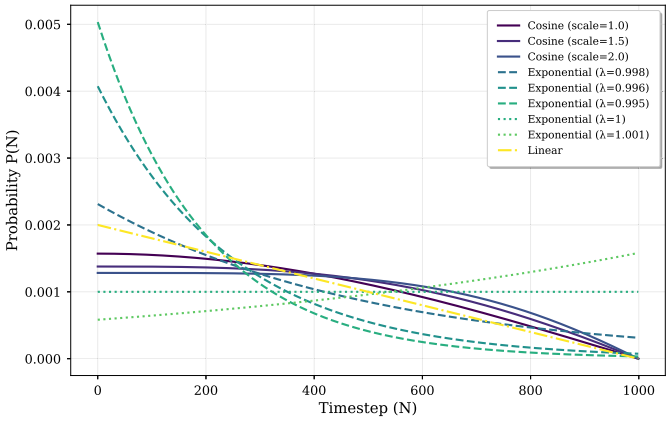

Fig. 7. Distribution of different time sampling functions.

From Table IV, it is evident that pretraining significantly enhances the model's performance in remote sensing super-resolution tasks. Specifically, the model trained without pretraining outperforms the zero-shot transfer results in terms of PSNR, FID, and LPIPS metrics, indicating that training directly on a small dataset can achieve some level of performance, but it remains limited. In contrast, the model pretrained on a large-scale dataset and fine-tuned achieves the best values for PSNR and LPIPS (33.142 and 0.0861, respectively), while maintaining a comparable FID score (22.156) to the model trained without pretraining. This demonstrates that pretraining not only improves reconstruction accuracy (PSNR) but also enhances the visual quality (LPIPS) and authenticity (FID) of the generated images.

3) Time Sampling Functions: The choice of time sampling functions during training has a significant impact on the results. As discussed in SD3 [69], using a concave function for uniform sampling between Gaussian and image distributions yields better results. In our method, since the prior distribution is given, we explore the effects of different time sampling functions, as shown in Fig. 7. For the cosine function in (10), we set α(scale) to 1.0, 1.5, and 2.0. For the exponential function in (11), we set λ to 0.998, 0.996, 0.995, 1, and 1.001. We also include a linear sampling function as a baseline for comparison. The experimental results are summarized in Table V.

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TABLE V
PERFORMANCE COMPARISON OF DIFFERENT TIME SAMPLING STRATEGIES

SAMPLING STRATEGY	PARAMETER SETTINGS	PSNR↑	FID↓	LPIPS↓
LINEAR	-	33.053	22.756	0.0918
COSINE	SCALE=1.0	33.043	22.738	0.0912
COSINE	SCALE=1.5	33.076	22.763	0.0915
COSINE	SCALE=2.0	33.093	22.768	0.0901
EXPONENTIAL	$\lambda = 1.001$	32.913	22.922	0.1022
EXPONENTIAL	$\lambda = 1.000$	33.113	22.731	0.0901
EXPONENTIAL	$\lambda = 0.998$	33.131	22.702	0.0895
EXPONENTIAL	$\lambda = 0.996$	33.138	22.688	0.0899
EXPONENTIAL	$\lambda = 0.995$	**33.142**	**22.682**	**0.0891**

TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT SEMANTIC GUIDANCE STRATEGIES

SEMANTIC STRATEGY	PSNR↑	FID↓	LPIPS↓
NONE REFERENCE	32.286	26.156	0.1463
RGB-IMAGE	×	×	×
LPIPS-VGG	33.108	22.934	0.1027
RESNET-18	33.026	22.881	0.0952
VIT-B/32	33.084	22.803	0.0938
CLIP-VIT-B/32	**33.142**	**22.682**	**0.0891**

From Table V, it is clear that different time sampling functions significantly affect model performance. The linear sampling function, as a baseline, achieves a PSNR of 33.053, an FID of 22.756, and an LPIPS of 0.0918, demonstrating stable but suboptimal performance. For the cosine function, the model achieves the best performance with scale = 2.0, yielding a PSNR of 33.093, an FID of 22.768, and an LPIPS of 0.0901, outperforming other parameter combinations. This indicates that larger scale values in the cosine function better balance the sampling distribution during training, thereby improving model performance. For the exponential function, the model achieves optimal performance when $\lambda = 0.995$ (PSNR = 33.142, FID = 22.682, and LPIPS = 0.0891), significantly outperforming other parameter settings. Notably, when $\lambda = 1.001$, the model performs the worst (PSNR = 32.913, FID = 22.922, and LPIPS = 0.1022), suggesting that excessively large λ values cause the sampling distribution to deviate from the optimal state, degrading model performance. As λ decreases from 1.001 to 0.995, the model's performance gradually improves, further validating the effectiveness of smaller λ values in optimizing the sampling distribution. Comparing the three sampling strategies, the exponential function with $\lambda = 0.995$ achieves the best performance, followed by the cosine function scale = 2.0, while the linear function performs relatively poorly. This demonstrates that in our method, using nonlinear sampling functions (especially the exponential function) can significantly enhance the model's super-resolution performance.

4) Impact of Semantic Guidance: Since flow matching loses information about the initial distribution during inference, semantic information can serve as an alternative form of the initial distribution to guide the sampling process. In this section, we discuss the impact of different semantic guidance strategies on the final results, including raw RGB image distribution, LPIPS semantic information, ResNet [70] semantic information, ViT [71] semantic information, and Clip semantic information. All the semantic features are extracted from officially pretrained models, and their parameters are frozen during training to avoid introducing additional training complexity. The experimental results are shown in Table VI. From Table VI, it is evident that semantic guidance strategies significantly affect model performance. Without semantic reference (None Reference), the model achieves a PSNR of 32.286, an FID of 26.156, and an LPIPS of 0.1463, indicating relatively poor performance. This suggests that the lack of semantic guidance causes the model to lose important contextual information during reconstruction, thereby degrading the quality of the generated images. For RGB image distribution, since it is difficult to directly use as semantic information in the flow matching framework, no effective results were obtained (marked as ×). In contrast, semantic guidance strategies based on pretrained models significantly improve model performance. Among them, LPIPS-VGG semantic information achieves a PSNR of 33.108, an FID of 22.934, and an LPIPS of 0.1027, outperforming the no-reference case but still lagging behind other semantic strategies. ResNet-18 semantic information further enhances model performance (PSNR = 33.026, FID = 22.881, and LPIPS = 0.0952), demonstrating its ability to effectively capture mid-level image features. ViT-B/32 semantic information further optimizes the results (PSNR = 33.084, FID = 22.803, and LPIPS = 0.0938), highlighting the advantage of the transformer architecture in modeling global features. Finally, Clip-ViT-B/32 semantic information achieves the best performance (PSNR = 33.142, FID = 22.682, and LPIPS = 0.0891), significantly outperforming other semantic strategies. This indicates that the semantic features learned by the Clip model through joint vision–language pretraining can better guide the image reconstruction process, particularly excelling in capturing high-level semantic information and global contextual relationships.

5) Loss Function Weights: In this section, we explore the impact of different Laplacian perceptual loss weights λ_l on

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TABLE VII
PERFORMANCE COMPARISON OF DIFFERENT LAPLACIAN PERCEPTUAL LOSS WEIGHTS

PARAMETER SETTINGS	PSNR↑	FID↓	LPIPS↓
$\lambda_l = 0$	**33.257**	25.553	0.2171
$\lambda_l = 0.01$	33.179	24.821	0.1924
$\lambda_l = 0.05$	33.137	22.796	0.1031
$\lambda_l = 0.2$	32.891	22.881	**0.0834**
$\lambda_l = 0.1$	33.142	**22.682**	0.0891

TABLE VIII
PERFORMANCE COMPARISON OF DIFFERENT SAMPLING STEPS DURING INFERENCE

SAMPLING SETTINGS	PSNR↑	FID↓	LPIPS↓
STEPS=1	33.127	22.693	0.0916
STEPS=2	33.135	22.690	0.0908
STEPS=3	33.139	22.685	0.0899
STEPS=5	**33.142**	**22.682**	**0.0891**

model performance. The experimental results are shown in Table VII. From Table VII, it is clear that the value of λ_l significantly affects model performance. When $\lambda_l = 0.01$, the model achieves a high PSNR (33.179) but performs poorly in FID (24.821) and LPIPS (0.1924), indicating that a small λ_l value preserves pixel-level accuracy but falls short in perceptual quality and image authenticity. As λ_l increases, the model's perceptual quality and image authenticity gradually improve. When $\lambda_l = 0.05$, the FID decreases to 22.796, and the LPIPS decreases to 0.1031, indicating that the Laplacian perceptual loss begins to take effect. When $\lambda_l = 0.1$, the model achieves the best balance across all three metrics (PSNR = 33.142, FID = 22.682, and LPIPS = 0.0891), demonstrating that this weight value effectively optimizes the model's detail reconstruction capability and perceptual quality. However, when λ_l increases further to 0.2, although the LPIPS decreases to 0.0834, the PSNR drops to 32.891, and the FID slightly increases to 22.881. This suggests that an excessively large λ_l value causes the model to overemphasize perceptual loss, sacrificing some pixel-level accuracy.

6) Impact of Sampling Steps: In this section, we discuss the impact of different sampling steps during the inference phase on the final results. The experimental results are summarized in Table VIII. From Table VIII, it is evident that as the number of sampling steps increases, the quality of the generated images improves. However, this also leads to a significant increase in inference time. Therefore, in practice, a tradeoff must be made between generation quality and inference efficiency.

VI. CONCLUSION

This article proposes an SfmSR for fast and accurate remote sensing image super-resolution reconstruction. By introducing a flow matching framework, SfmSR significantly reduces the number of sampling steps while maintaining high-quality reconstruction. The experimental results demonstrate that SfmSR outperforms the existing methods on multiple remote sensing image datasets, particularly excelling in key metrics such as PSNR, FID, and LPIPS. Compared with traditional diffusion models that require 20–1000 sampling steps, SfmSR achieves better results in fewer than five steps. In addition, SfmSR demonstrates significant advantages in model complexity and inference efficiency. It achieves fast inference with fewer parameters and lower memory usage, meeting the real-time requirements of practical applications. Compared with the existing SOTA methods, SfmSR not only reaches new heights in reconstruction quality but also exhibits clear advantages in inference speed and computational efficiency. Future research could focus on further optimizing semantic guidance strategies, exploring the impact of additional types of semantic information on model performance, applying consistency models to develop more effective one-step sampling methods, and validating their generalization capabilities in other downstream tasks.

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